Summary of Research and Future Research Plans for Master's Program

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Program to be enrolled in: Master's Program in Materials Innovation

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1. Statement of Financial Resources

Please indicate below your source and amount of funding for your study at University of Tsukuba.

Sources of Funds	Amount (Japanese yen)
Personal savings	¥ 0
Parent or relative （Please specify the relation: Mother and Father ）	¥ 10.000.000
Government / sponsoring agency （The name of your sponsor: Japanese Government ） MEXT Scholarship from September 2023-August 2027 ） * Please attach a copy of scholarship award letter.	¥ 5.760.000
Others （Please specify the details: ） ）	¥ 0
TOTAL:	¥ 15.760.000

I hereby certify that all information on this statement is true and accurate and that the stated funds are available for my educational expenses at University of Tsukuba.

Student's signature _____ Date 2026.09.02

2. [Research Status up to Date and Future Research Plans in *three* pages] (Change/addition of the form is not allowed. (The same shall apply hereafter))

(1) Describe research status up to date if you have performed research so far. The background, purposes, methods, characteristic points, major results, and conclusions should be written clearly. You can include figures, tables, and schematics.

(2) Describe future research plans containing background, purposes, methods, and points to be solved. You can include figures, tables, and schematics.

1.Title: Design of MgH_2–Ni-2D Composites Using Nickel Complex Precursors and Two-Dimensional Nanosheet Supports for Enhanced Hydrogen Release and Storage Property

2.Background

Hydrogen is a promising clean-energy carrier due to releasing only H_2O(water) during combustion/fuel cell reaction instead of greenhouse gases such as CO_2. Solid-state hydrogen storage using metal hydrides offers high gravimetric and volumetric densities, improved safety, and potential economic advantages [1,2,3]. MgH_2 is particularly attractive because magnesium is abundant and MgH_2 has a theoretical hydrogen capacity of 7.6 wt.%, a volumetric capacity of approximately 110 kg m^{-3}, and a stable formation enthalpy of about −74.5 kJ mol^{-1}. [1,2,4] However, strong Mg–H bonding causes slow hydrogen absorption and desorption, hydrogen-release temperatures above 300 °C, and limited cycling stability. [1,2] These limitations can be mitigated through nano structuring, Ni catalysts, and two-dimensional (2D) materials.[2] Nano structuring improves hydrogen diffusion; Ni is a **low-cost and effective catalyst for Mg/MgH_2**, reducing the activation barrier and significantly enhancing hydrogen absorption and desorption kinetics; while 2D materials suppress particle agglomeration and improve cycling performance. [2,.5] For example, Shen et al. reported that an MgH_2–$Ni(acac)_2$–expanded graphite composite released 7.03 wt.% H_2 after 10 cycles.[5] Previous studies have shown that the catalytic performance depends not only on the amount of Ni but also on its chemical environment. The atoms and ligands coordinated to Ni may influence electron transfer, the stability of Ni–H and Mg–H bonds, the formation of Mg–Ni phases, and the resulting hydrogen-storage behaviour. Zhang et. al 2017 reported the electronegativity of the element bonded to Ni in a nickel-based compound doped into MgH_2 affects the hydrogen release/storage kinetics and performance of MgH_2. As Ni binds with the low-electronegativity element, the stability of the Ni-H bond decrease and the bond between Ni-H bond is easier to break, promoting dehydrogenation. [6] Likewise, the charge and electron transfer of Mg also matters for hydrogen absorption/release kinetics of MgH_2 composites. During dehydrogenation, efficient transfer of electrons from H^- toward Mg^{2+} may promote Mg–H cleavage, whereas during hydrogenation, electron transfer from Mg toward dissociated hydrogen may facilitate MgH_2 formation. For quicker hydrogen H^- release, Mg should be more negative. Conversely, for quicker hydrogen absorption, Mg should be more positive. [7,8,9] The ligand environment of the Ni precursor may regulate interfacial electron transfer during MgH_2 dehydrogenation and Mg hydrogenation.

However, influence of the chemical environment and **nickel complex as Ni catalyst precursor** on hydrogen desorption and absorption of MgH_2 has not yet been fully clarified. Previous studies have primarily investigated individual Ni-based catalyst nickel complexes precursors for improving MgH_2 hydrogen storage properties. However, differences in Ni loading (different nickel complex wt%. in MgH_2-nickel complex), milling conditions, supports, and characterisation make it difficult to isolate the effect of precursor chemistry.

3.Purpose

By systematically comparing different nickel complex precursors under the same ball milling parameters and controlled preparation conditions, this study aims to 1) improve the comparison of each nickel complex doped-MgH_2 composite's dehydrogenation performance and 2) clarify the relationships among precursors. Therefore, this study will compare multiple nickel complex precursors under controlled conditions and correlate hydrogen release and storage behaviour with precursor decomposition, Ni/Mg chemical states, catalyst dispersion, interfacial contact, and Mg–Ni phase formation. We want to investigate the role of the interaction between ligand/nearest element of Ni which affects the charge state of Ni and Mg, ligand decomposition, Ni dispersion, MgH_2–Ni interfacial contact, Mg–Ni phase formation, and hydrogen-release behaviour. **The objective of this research is therefore to clarify how the ligand environment of nickel complex precursors affects the hydrogen release and storage properties of MgH_2.** Nickel acetylacetonate, $Ni(acac)_2$, was first selected as a model precursor to optimize the ball-milling conditions of MgH_2+$Ni(acac)_2$ due to the high performance of the composite[5]. Candidate precursors for later comparison include $NiCl_2 \cdot 6H_2O$, $Ni(OH)_2$ and nickel phthalocyanine. For each nickel complex, the interaction between Ni and its nearest neighbour element/compound or ligand will be considered. After identifying a suitable Ni precursor, hydrogen boride sheets or graphene will be incorporated as two-dimensional supports to suppress particle agglomeration and improve cycling stability.

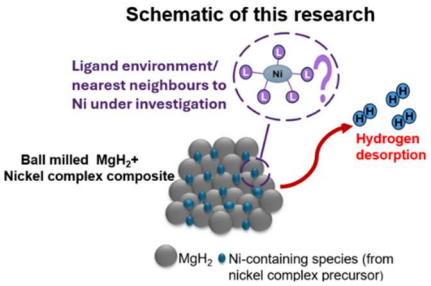


Figure 1 Schematic of this research: Investigating ligand environment around Ni in MgH_2-Ni complex composite precursor to uncover its role in hydrogen release and storage property

Two-dimensional nanosheets can improve the hydrogen-storage performance of MgH_2 by suppressing particle agglomeration and maintaining active interfaces. For example, Zhang et al. [10] produced highly dispersed MgH_2 nanoparticles on graphene, where nanoscale confinement enhanced hydrogen-desorption kinetics. Hydrogen boride (HB), synthesised in my laboratory, is another promising support because of its ultrathin structure and theoretical hydrogen capacity of 8.5 wt.%. [11,12] In MgH_2–Ni/HB composites, Ni promotes hydrogen transfer and recombination, while HB helps maintain the dispersion of Ni and MgH_2. Noguchi [1] reported that MgH_2–Ni/HB containing only 2 wt.% Ni released hydrogen at approximately 550 K, about 50 K lower than ball-milled MgH_2, demonstrating improved hydrogen-release kinetics with a low catalyst loading. MgH_2-Ni interface contact is a key for Mg nucleation, which makes the Mg/MgH_2 dehydrogenation reaction starts sooner and the overall desorption rate increases.

Our hypothesis is that ball-milled MgH_2–Ni composites will exhibit lower hydrogen desorption temperatures, faster dehydrogenation kinetics, improved cycling stability, and better hydrogen storage performance than commercial and pure ball-milled MgH_2. The surrounding ligand in a nickel complex will affect the multivalence state of Ni, charge of Mg, bonding of Ni-H and affect hydrogen release/storage[6]. Furthermore, the addition of graphene or HB is expected to prevent agglomeration of MgH_2 and Ni particles, thereby further enhancing cycle stability.

Characterisation techniques

Method	Function/Overview
XRD	Crystallinity and phase identification
IR	Chemical bonding
SEM-EDX and TEM	Morphology
XPS	Surface chemical states of Ni and Mg, and interfacial bonding
TG–DTA-MS and TPD	H_2 release temperature and activation energy
PCT	H_2 storage capacity

Table 1: Characterisation techniques used in this research.

4.Methodology(Table 1): The list of characterisation methods used for this research is summarised in Table 1.

- **XRD** identifies crystalline phases and evaluates crystallite-size changes after ball milling and hydrogen desorption.
- **IR spectroscopy** examines chemical bonding and changes in the nickel-complex ligands during sample preparation and heating.
- **SEM-EDX and TEM** observes particle morphology, size, dispersion, and agglomeration.
- **XPS** determines the surface chemical states of Ni and Mg and investigates possible interfacial bonding with HB or graphene.
- **TG–DTA–MS and TPD** determine the hydrogen-release temperature, thermal behaviour, released gas species, and apparent desorption

(continued)

activation energy.

- **PCT measurements** evaluate hydrogen-storage capacity, absorption–desorption behaviour, and reversibility.

5. Research Status Up to Date

Part 1: B3-M1 Investigation of MgH_2–Nickel Complex Composite Hydrogen Release/Dehydrogenation Property

MgH_2 will be ball-milled with different nickel complexes to obtain a nanocrystalline size, improve Ni dispersion, and determine how the ligand environment influences hydrogen-release behaviour. The main targets are lower hydrogen-desorption temperatures and reduced apparent activation energy compared with ball-milled MgH_2.

Step 1: B3 2nd semester : Ball-Milling Optimisation and Dehydrogenation of $Ni(acac)_2$-Derived MgH_2–Ni Composites

$Ni(acac)_2$ is used as the initial model precursor because of its reported effectiveness in MgH_2-based systems[5]. Intensive one-step milling and gentle two-step milling are compared to identify reproducible conditions for future nickel complexes while examining whether preservation of the precursor structure affects performance. The experimental workflow includes pre-characterisation, ball milling, post-milling characterisation, hydrogen desorption measurements using TG–DTA and temperature-programmed desorption (TPD), and post-desorption characterisation, with X-ray diffraction (XRD) used to evaluate crystallinity. TG–DTA, TPD, and XRD measurements have been completed, while IR analysis before and after dehydrogenation remains to be conducted.

5.1 Experiment

1. Sample preparation

First, commercial MgH_2 was ball-milled with seven balls at 400 rpm, using a ball-to-sample mass ratio of 48:1, for 4 hours (44 cycles) in a Fritsch planetary ball mill. For all samples, each cycle consisted of 5 minutes of rotation followed by a 30-second rest period to minimise temperature rise during prolonged milling. All milling and handling of the powders were performed inside an argon-filled glovebox. We synthesised MgH_2–$Ni(acac)_2$ composites by ball milling commercial MgH_2 and $Ni(acac)_2$ using two methods: (1) a one-step "intensive" ball-milling process, and (2) a two-step "gentle" process, where MgH_2 was first milled before gentle mixing with $Ni(acac)_2$ to preserve its structure.

Intensive method: l MgH_2 containing 2 wt.% $Ni(acac)_2$ was ball-milled together at 400 rpm for 4 h and 44 cycles under Ar using four balls and a ball-to-sample mass ratio of 48:1.

Gentle method: First, MgH_2 was first ball-milled alone under the same conditions as the intensive method for 4 h and with 4 ball, 400 rpm, and 44 cycles. Then, 2 wt.% $Ni(acac)_2$ was added and milled for a further 2 h, 24 cycles and this time at 250 rpm and to better preserve the $Ni(acac)_2$ structure. The reason we used less balls for ball milling MgH_2-$Ni(acac)_2$ is to decrease sample amount needed to balance ball:sample masss ratio = 48:1.

2.TG-DTA-QMS and TPD

TG–DTA was combined with hydrogen temperature-programmed desorption (H_2-TPD) using a STA-2500 Regulus analyser under argon flow. The evolved gases were transferred through a capillary to a quadrupole mass spectrometer and monitored. Three samples were analysed: ball-milled MgH_2, intensively ball-milled MgH_2–$Ni(acac)_2$, and gently ball-milled MgH_2–$Ni(acac)_2$, with sample masses of 10.9, 9.7, and 8.8 mg, respectively. Hydrogen calibration was performed at room temperature by injecting H_2 volumes 1) for MgH_2 0.01,0.05, 0.1, 0.2 and 0.5 mL, while 2) for MgH_2-$Ni(acac)_2$ composites 0.01, 0.05, 0.1 and 0.5 mL. All samples were heated from 25 to 600 °C at 10 K min^{-1}. The TG–DTA data were recorded using NETZSCH software and subsequently processed in Excel to obtain the H_2-TPD profile.

3.XRD

XRD measurements were performed at approximately 25°C using a Rigaku MiniFlex diffractometer with Cu Kα radiation (40 kV, 15 mA) to identify sample phases. Data were collected over a 2θ range of 5–90° with a step size of 0.0100° and speed of 5.0 ° using SMARTLab software. Samples were sealed in a Rigaku atmosphere separator with Kapton film to minimize air exposure, and most transfers were conducted in an argon glovebox. The samples we measured were commercial MgH_2, ball milled MgH_2 (4 balls), ball milled MgH_2(7 balls) after dehydrogenation, and intensively vs gently ball milled MgH_2 +2 wt.% $Ni(acac)_2$ before and after dehydrogenation. However, the ball-milled MgH_2 and gently ball-milled MgH_2–2 wt.% $Ni(acac)_2$ samples measured after TG–DTA were transferred in air because of limited sample quantities.

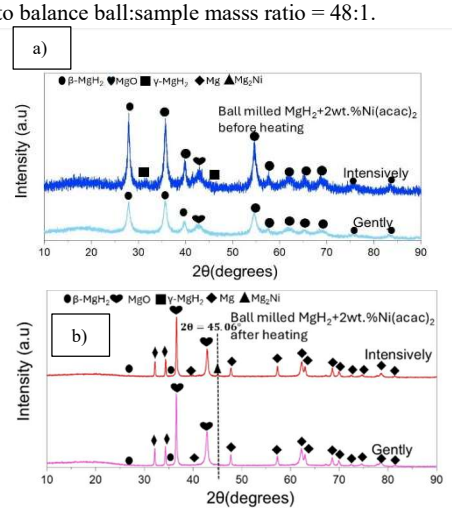


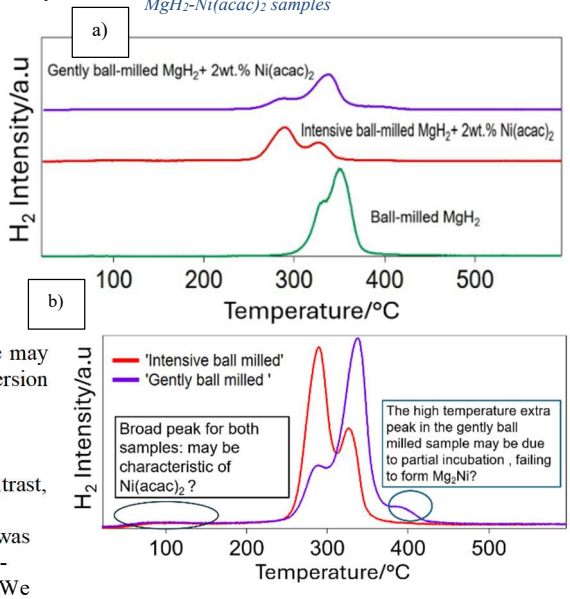
Figure 2 XRD results of gently and intensively ball milled MgH_2+ 2 wt.% $Ni(acac)_2$ composite before and after TG-DTA (dehydrogenation)

5.2 Result

XRD

By systematically comparing different nickel complex precursors under controlled preparation conditions, this study aims to clarify the relationships among precursors. Figure 2 compares the XRD patterns of intensively and gently ball-milled samples before and after dehydrogenation. No crystalline $Ni(acac)_2$ peaks were detected after milling, suggesting decomposition, reaction with MgH_2, or fine dispersion below the XRD detection limit[5]. Before dehydrogenation, both composites mainly contained β-MgH_2 and minor MgO, while metastable γ-MgH_2 appeared only in the intensively milled sample. After heating, Mg and MgO were dominant. A weak "hydrogen pump" Mg_2Ni peak was observed in the intensively milled sample at 2θ=45.06° but not in the gently milled sample, indicating that milling conditions influenced the reaction between Ni species and MgH_2.(Figure 3b) Before dehydrogenation, intensively ball-milled MgH_2+$Ni(acac)_2$ had the largest average crystallite size (8.859 nm), the gently milled composite (8.458 nm), while ball-milled MgH_2 had the smallest value (8.251 nm), calculated from Scherrer Equation (shape factor k=0.89, $\lambda = 0.154nm$). After dehydrogenation, the gently milled composite showed substantial grain growth to 41.856 nm, whereas the intensively milled composite retained a much smaller crystallite size of 9.426 nm. This difference may result from the higher defect density, residual microstrain, and finer Ni-species dispersion produced by intensive milling, which suppressed crystallite growth.

Figure 3 a) Hydrogen release profile of gently vs intensively ball milled MgH_2+2 wt.% $Ni(acac)_2$ and ball milled MgH_2 b) Side by side release profile of both MgH_2-$Ni(acac)_2$ samples



TPD

The TPD hydrogen release profile of ball milled MgH_2, intensively ball milled MgH_2+$Ni(acac)_2$ and gently ball milled MgH_2+$Ni(acac)_2$ is plotted in Figure 3a. By contrast, commercial MgH_2 with the same TG-DTA heating speed of 10°C/min(=10K/min) has a sharp dehydrogenation peak at 440°C. [13] The onset temperature of ball milled MgH_2 was approximately 310 °C and the highest dehydrogenation peak at 350.3°C. Both $Ni(acac)_2$-containing samples have a broad low temperature peak at around 66-133°C(Figure 3b). We will confirm whether this is a characteristic of $Ni(acac)_2$ by conducting TG-DTA and TPD on $Ni(acac)_2$ by itself. Both $Ni(acac)_2$-containing samples began releasing hydrogen at lower temperatures, the gently ball milled sample at 257.86 ℃ and intensively ball milled sample at 260.43 ℃.

(continued)

Interestingly, even though only a small amount of $Ni(acac)_2$ (only 2wt.%) was added and mixed to MgH_2 during ball milling, there was an apparent decrease in temperature in comparison with commercial and pure ball milled MgH_2. This may be due to the breaking of $Ni(acac)_2$ in ball-milling, causing finely dispersed Ni among the MgH_2-Ni complex and improving the contact of Mg with Ni particles. We will verify the Ni dispersion through TEM. The intensively milled sample exhibited two principal hydrogen-release peaks at approximately 288.27 and 329.52 °C. In comparison, the gently milled sample exhibited a more complicated multistep profile, with peaks near 284℃, the main peak at 337.271 ℃ and an additional broad high-temperature feature centred at approximately 393 °C (Figure 3a-b). The latter peak may originate from MgH_2 regions that were insufficiently activated by the Ni-derived catalyst because of heterogeneous catalyst dispersion, limited MgH_2–Ni interfacial contact, or an insufficient number of accessible catalytic sites. The inability of $Ni(acac)_2$ in the gently ball milled composite to generate a detectable Mg_2Ni phase (as shown by XRD result) may also have limited the establishment of an efficient Mg–Ni-mediated hydrogen-transfer, which may lead to partial incubation. (Figure 3b) The intensively milled MgH_2–$Ni(acac)_2$ composite showed the most favourable hydrogen-release behaviour. Its main desorption peak occurred at a substantially lower temperature than that of both commercial and ball-milled MgH_2. The possible formation of Mg_2Ni may have contributed to this improvement by establishing an Mg–Ni-mediated hydrogen-transfer pathway or "hydrogen-pump" effect. [5,14, 15] Nevertheless, additional experiments are required before directly correlating Mg_2Ni formation with the measured desorption kinetics. The results demonstrate that $Ni(acac)_2$ effectively lowers the hydrogen-release temperature of MgH_2 and that its effectiveness depends on the ball-milling milling procedure. The intensively milled MgH_2–$Ni(acac)_2$ composite showed the most favourable hydrogen-release behaviour. Its main desorption peak occurred at a substantially lower temperature than that of ball-milled MgH_2. Therefore, the intensive method is currently the most suitable standard condition for comparing additional nickel complex precursors. The current findings also show that evaluating only the initial hydrogen-release temperature is insufficient. Phase formation, catalyst dispersion, MgH_2–Ni interfacial contact, the origin of the multistep hydrogen release, and structural evolution during heating must also be considered when identifying the optimum catalyst system. **These points must be resolved before different nickel complex precursors can be compared reliably.**

6.Future Research Plan

Part 1 continued

Step 2: B3-B4 1st semester Detailed Characterisation of MgH_2–$Ni(acac)_2$

Further analysis will clarify the origins of the hydrogen-release peaks and the effect of milling conditions on the catalytic Ni species from $Ni(acac)_2$ precursor. TG–DTA–QMS of pure $Ni(acac)_2$, hydrogen calibration, baseline correction, and peak deconvolution will be used to identify evolved gases and quantify gravimetric hydrogen release(wt %). IR, XPS, and SEM–EDX will examine bonding, chemical states, morphology, and elemental distribution before and after dehydrogenation. TEM will be conducted to examine the Ni dispersion on MgH_2-Ni composite.

Step 3: B4 1st semester-M1 1st semester Comparison with Other Nickel Complexes for Hydrogen Release Performance of MgH_2-NiX composite

After optimizing the procedure, the same workflow with the same experimental parameters will be applied to $NiCl_2 \cdot 6H_2O$, $Ni(OH)_2$, nickel phthalocyanine, nickelocene, and other candidates. The same ball milling conditions will be applied to each ball milled MgH_2 and nickel complex composites. Each ball milled composite will have MgH_2-2 wt.% NiX composition, to ensure the same doping ratio of each complex. Similarly, the same TG-DTA (heating) conditions will be conducted on each MgH_2-NiX composite Their hydrogen-release temperature, hydrogen capacity, reaction steps, and crystalline phases will be compared to determine how precursor structure and decomposition products affect Ni-phase formation, MgH_2–Ni contact, and dehydrogenation kinetics. To determine the activation energy, different heating rates (such as 5K/min and 20K/min) will also be tested in the TG-DTA of the composites, and the apparent dehydrogenation activation energy will be estimated using an appropriate kinetic analysis such as the Kissinger method.

Step 4: B4-M1 2nd semester Elucidation of the Hydrogen-Release Mechanism and the Ligand Environment Influence

The results will be used to determine whether catalytic performance is mainly controlled by the atoms coordinated to Ni, precursor decomposition pathways, Ni-species dispersion, or Mg–Ni phase formation. Possible inhibiting effects, including those caused by chlorine-containing species[16], will also be investigated to clarify the reaction pathways and rate-controlling factors.

From part 1, the most promising MgH_2–nickel complex composite will be selected based on its ability to lower the hydrogen-release onset and peak temperatures, reduce the apparent desorption activation energy, release a large amount of H_2 without a substantial high-temperature component, and highest gravimetric H_2 release.

Part 2: M1 1st semester Hydrogen Storage and Cycling Performance

The most promising nickel precursor will be evaluated by pressure–composition–temperature(PCT) measurements to determine reversible hydrogen capacity, equilibrium pressure, absorption–desorption behaviour, and thermodynamic properties. Repeated hydrogenation and dehydrogenation tests will then be performed to assess cycling stability.

Part 3: M1-M2 Addition of 2D Nanosheet HB/graphene to the MgH_2-Ni composite to Improve Kinetics and Cyclability

In the final stage, HB or graphene will be introduced into the best-performing MgH_2-Ni composite to improve kinetics and cyclability by aiming to suppress Mg/MgH_2 particle growth, maintain Ni dispersion, and active Mg-Ni interfaces, and improve hydrogen diffusion and cycling stability. MgH_2, MgH_2-Ni, and MgH_2-Ni-2D nanosheet composites will be compared under identical conditions. We will use XRD and IR for characterisation, TG-DTA-QMS and TPD for dehydrogenation investigation, and PCT for hydrogen absorption-desorption behaviour. We will use SEM-EDX to evaluate morphology, agglomeration and elemental distribution and XPS to investigate changes in the chemical states and interactions in chemical states. For HB-containing samples, possible Mg-B or Ni-B interactions will be investigated, while for graphene-containing samples the possible Mg-C or Ni-C interactions will be investigated. Other promising 2D support material, such as MXene, may also be a candidate.

7. Points to be Solved

This research aims to boost the dehydrogenation performance of MgH_2, a highly promising hydrogen storage material, by adding nickel complexes as Ni catalyst precursor through ball milling. Previous research often compares only individual nickel complex-doped MgH_2 , and each paper uses different ball milling conditions, different wt.% of nickel complex in the MgH_2-Ni composite, and different dehydrogenation measurement and different hydrogen absorption measurement conditions. We will use the same ball milling conditions and controlled experimental conditions **to better compare the effect of different nickel complexes and their ligand influence alone**, **without other factors** such as ball milling parameter influence, Ni loading, and TG-DTA and PCT conditions. We wish to improve the charge state of Ni and Mg, which will affect the strength and stability of Ni-H and Mg-H bonds, to boost the dehydrogenation kinetics of MgH_2-Nickel complex composites and elucidate the catalytic reaction pathways. The best performing MgH_2-nickel composite will then be subsequently evaluated for hydrogen storage using PCT. Finally, graphene or HB will be incorporated into the best-performing MgH_2–Ni composite to further improve hydrogen storage performance and cycling stability. This will path a comprehensive nano catalyst design guideline into efficient hydrogen release/storage of MgH_2, a promising and low -cost hydrogen storage metal hydride with abundant Mg, and help form a cleaner, more sustainable future with hydrogen energy source.

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3. Reasons for Selecting Host Researcher

Describe the reasons for selecting a host researcher. You need to contact host researcher prior to this application.
What made the applicant know the host researcher, and the status of arrangement regarding research implementation after the acceptance?

Developing environmentally friendly and renewable energy technologies is my dream and passion. I dream of the world where humans live in harmony with the environment, where everything is sustainable. Materials science plays a crucial role in this field, as materials are the fundamentals of advancing technologies. I am especially interested in hydrogen storage since, since hydrogen is environmentally friendly as an energy source (such as hydrogen fuel cell and electrolysis) because the product released after the fuel cell reaction/combustion is water. In a freshman course Introduction to Interdisciplinary Engineering II, Professor Kondo introduced his lab and on surface chemistry. I was deeply impressed by the research done in his lab, since his lab focuses on environmentally friendly materials especially on solid-state hydrogen storage and electrochemistry. Thus, I applied to enter Kondo-Oki laboratory to learn research in hydrogen storage materials. Furthermore, Kondo Lab is the first in the world to synthesise a novel 2D material hydrogen boride (HB). HB is a promising material which can promote hydrogen storage and dehydrogenation kinetics. The potential of HB application expands beyond energy/environmental applications, such as antimicrobial effect. Thus, the research of Kondo-Oki Lab aims for the betterment of both the environment and human society. The research in Kondo-Oki Lab is rapid, breakthrough and innovative.

Based on this interest, I joined the Kondo–Oki Laboratory in my third undergraduate year, with Professor Kondo as my research supervisor. Professor Kondo introduced me to research about metal hydrides and ball milling and suggested my research topic, while a senior laboratory member proposes my research project and gives detailed guidance on the research theme, materials and the experimental procedures throughout the project. Since then, under my senior's guidance on metal hydrides, catalytic hydrogen-storage materials, ball-milling processes, and surface-characterization methods, I have pursued my strong interest in solid-state hydrogen storage by investigating the improvement of hydrogen-release kinetics in MgH_2 using different nickel complexes for my graduation research. For my research, we will test several nickel complexes as a catalyst precursor with MgH_2 to see optimal condition. As current condition shows, thoroughly investigating only one nickel complex $Ni(acac)_2$ will take time to see results, including changing different parameters and investigating reaction mechanisms. This deepens my interest, and I am excited to continue my research to compare many different complexes since each of them have different characteristics that can either boost or inhibit hydrogen storage. Being in Kondo-Oki Lab makes me realise how much I want to learn how to do research, especially regarding catalysts and surface chemistry.

Professor Kondo has kindly agreed to be my host researcher and supervise the proposed research which is continuation of my undergraduate research after acceptance. I want to continue my undergraduate research in graduate school to test more different promising combinations of MgH_2-NiX-2D nanosheet materials, and to understand catalytic mechanisms regarding hydrogen release/storage of the composites more deeply, specifically regarding the role of Ni ligands and Mg interactions in dehydrogenation and hydrogen storage kinetics. This will provide the necessary foundations to become a researcher with wide perspective and a deep understanding of materials science and catalytic mechanisms. I want to continue to graduate school to learn surface chemistry and catalyst science. Kondo-Oki Lab provides the perfect environment for students to learn how to advance new materials and their application into green energy generation and storage.